



ME 300:
Engineering Thermodynamics

Conservation of Mass

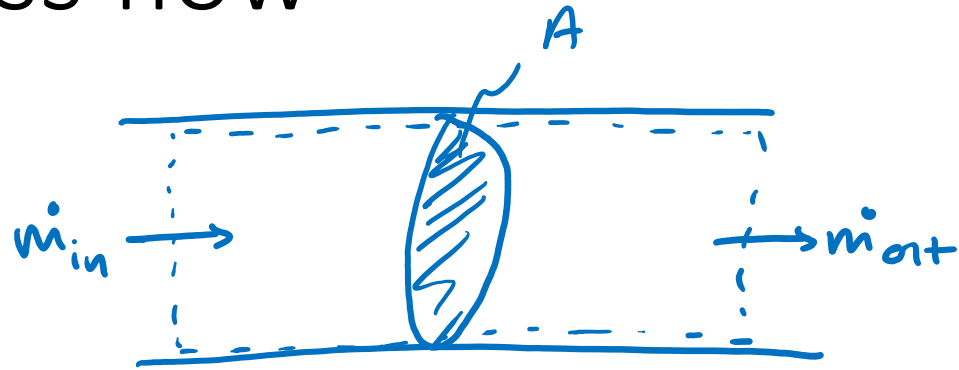
Mass conservation

Finite time: $\Delta M_{\text{sys}} = M(t_2) - M(t_1) = M_{\text{in}} - M_{\text{out}} + \cancel{M_{\text{gen}}}^0$

→ system: $\Delta M_{\text{sys}} = 0$

Instantaneous: $\frac{dM}{dt}_{\text{sys}} = 0$, $\underline{\underline{\frac{dM}{dt}_w = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} + \cancel{\dot{m}_{\text{gen}}}^0}}$

Mass flow

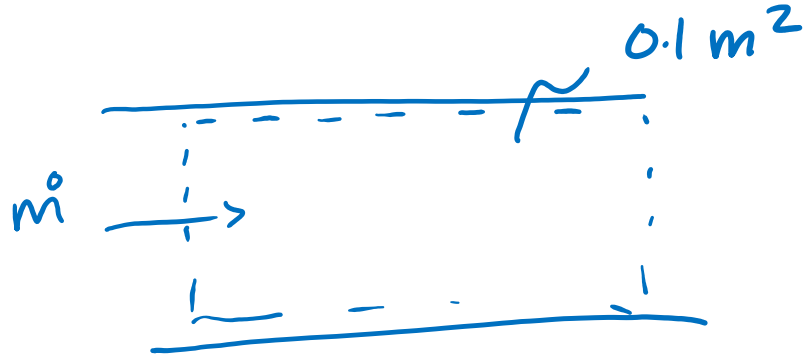


$$\dot{m} = \rho v A$$

$$\uparrow v_{av} = \frac{1}{A} \int_A v dA$$

$$\Rightarrow \frac{dm_c}{dt} = \dot{m}_{in} - \dot{m}_{out}$$

Mass flow example



$$\dot{m} = 10 \text{ kg/s}$$

$$\rho = 1000 \text{ kg/m}^3$$

$$v = ?$$

$$\boxed{\dot{m} = \rho v A}$$
$$v = \frac{\dot{m}}{\rho A}$$

$$= \frac{10}{(1000)(0.1)} \rightarrow v = 0.1 \text{ m/s}$$

$$\rho = \text{const}$$

$$\dot{m} = \rho v A$$

$$\dot{m} \uparrow, v \uparrow$$



For a given density, if the mass flow increases, does velocity increase, decrease or stay the same?

- A. Increase
- B. Decrease
- C. Stay the same